

ANALYZING JOB MARKET DATA WITH LLMS IN SHEETS AND AI REPORTING AGENTS

❖ INTRODUCTION

This project leverages Large Language Models (LLMs) integrated directly within Google Sheets to systematically extract structured information - including salary estimates, experience requirements, and programming language specifications - from hundreds of unstructured data analyst job postings, followed by automated analysis and visualization using AI Julius.

The dataset contains 400 job postings with information about salaries, experience requirements, and programming language requirements, enabling a comprehensive examination of how these factors relate to compensation in the data analyst job market.

❖ ANALYSIS RESULTS

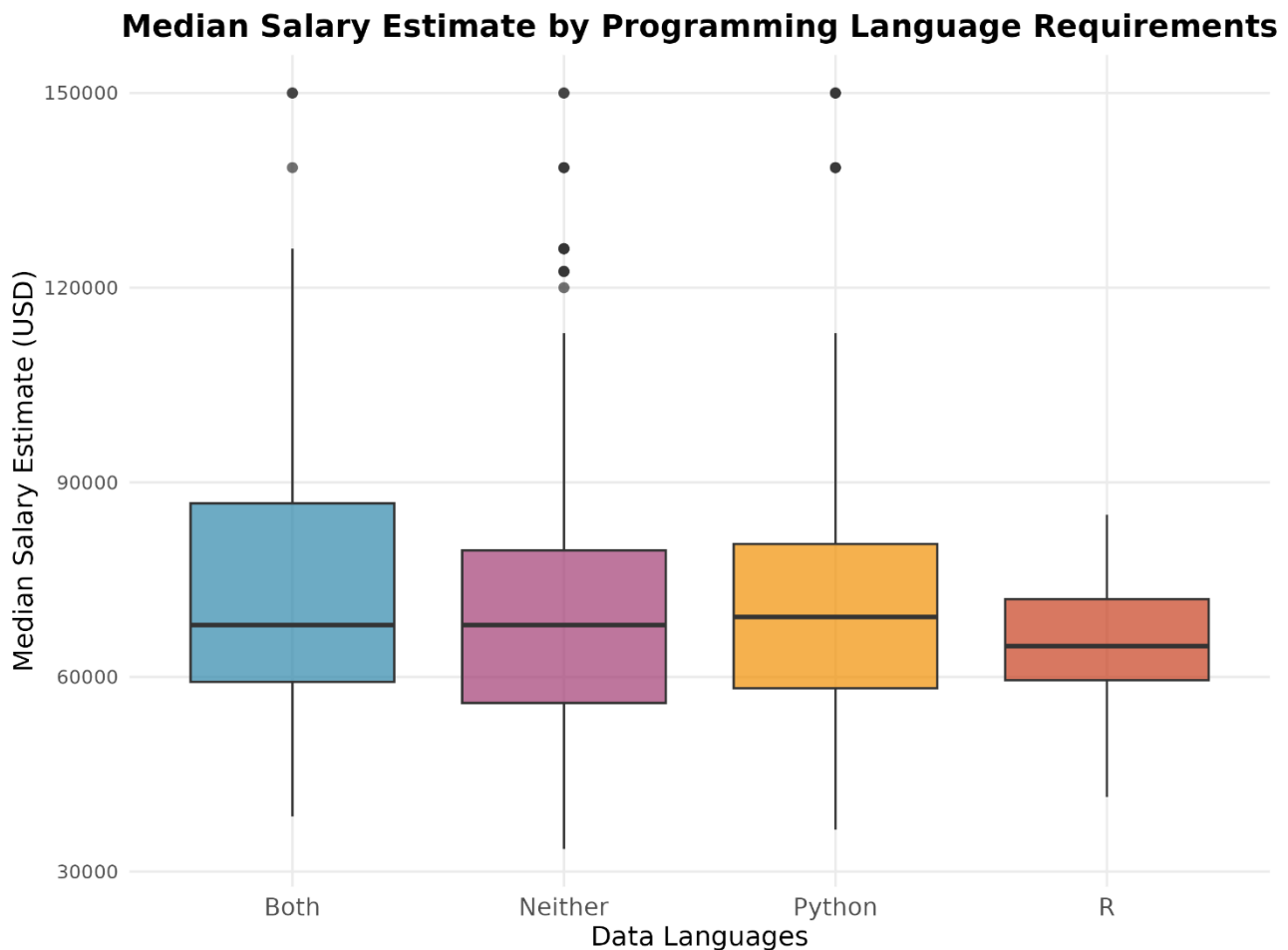
▪ Years of experience analysis



The analysis includes 309 observations (91 had missing experience data).

The scatter plot reveals a weak positive relationship between years of experience and median salary, with considerable variation across experience levels. However, given the small sample size and wide salary dispersion, this trend should be interpreted cautiously, as it may not fully represent the broader job market.

▪ Programming language requirements analysis



Jobs requiring both R and Python show slightly higher median salaries (\$68,000) compared to those requiring neither language, while R-only positions show the lowest median (\$64,750). However, the small sample sizes for R (n=14) and Python (n=58) categories limit the generalizability of these findings, and the overlapping salary ranges suggest that language requirements may not be the primary determinant of salary.

❖ MY REFLECTION

I found this assignment very interesting and engaging, and I developed new skills in both Sheets and Julius. I understand that I have only begun to explore the potential of generative AI, and each lesson helps me strengthen and expand my (nearly ground zero!) knowledge.